

Vishesh Garg

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Profile Summary

Computer Engineering undergraduate and full-stack developer with strong fundamentals in data structures, algorithms, and system design. Experienced building scalable, reliable backend systems and transactional applications using TypeScript, Python, Node.js, FastAPI, PostgreSQL, Redis, and Docker. Hands-on with REST API design, payment integration, data integrity under concurrency, CI/CD, and automated testing, with a focus on correctness and performance.

Education

Thapar Institute of Engineering and Technology (TIET)

Patiala, India

Bachelor of Engineering in Computer Engineering

2023 – 2027

- CGPA: 9.05/10 | Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Technical Skills

- **Languages:** TypeScript, JavaScript, Python, C++, SQL
- **Frontend:** React, Next.js, Zustand, Tailwind CSS, Vite
- **Backend:** Node.js, Express.js, FastAPI, Flask, REST APIs
- **Databases & Real-time:** PostgreSQL, Redis, MongoDB, Socket.IO
- **DevOps, Cloud & Testing:** Docker, GitHub Actions (CI/CD), Render, Vercel, Git, Jest, Pytest
- **AI/LLM Tools:** Google Gemini API, Prompt Engineering, Web Speech API

Projects

TicketVault – High-Concurrency Event Booking Platform

Apr 2026 – Jun 2026

TypeScript, Node.js, Express, PostgreSQL, Redis, React, Docker, GitHub Actions

- Built a full-stack ticket booking platform sustaining **550 req/s** at **p99 = 155 ms** under load; verified **zero double-bookings** in PostgreSQL with **500 concurrent users** contending for 50 seats.
- Guaranteed correctness under contention via a three-layer concurrency gate: atomic Redis Lua inventory decrement, PostgreSQL `SELECT FOR UPDATE NOWAIT` row locks (clean HTTP 423 instead of 500s), and an in-transaction status recheck.
- Implemented Redis-backed idempotency keys, per-endpoint rate limiting, JWT auth with single-use refresh-token rotation, and a sold-out waitlist queue with position tracking.
- Delivered real-time seat updates over Socket.IO (Redis pub/sub) with GitHub Actions CI running 25 Jest unit, integration, and end-to-end tests on every push.

Intervuo – AI-Powered Interview Simulation Platform

Aug 2025 – Oct 2025

Next.js, TypeScript, FastAPI, PostgreSQL, Google Gemini API, Pytest, Docker

- Built a full-stack AI mock-interview platform with a **Next.js 14 (App Router) + TypeScript** frontend and an async **FastAPI + PostgreSQL** backend, structured so core logic stays framework-independent and unit-testable.
- Integrated **Google Gemini API** for resume-grounded question generation and automated answer scoring using JSON mode for reliable parsing, with per-user rate limiting and timeouts on all AI endpoints.
- Built a client-side multimodal feedback pipeline using the Web Speech API (speech-to-text/text-to-speech) and MediaPipe (eye-contact tracking), keeping audio and video on-device and the backend cost-free.
- Managed auth and sessions with NextAuth + JWT and Alembic-versioned schemas; wrote Pytest suites and ran lint, type-check, and Docker image builds via GitHub Actions CI.

OrderFlow – Real-Time Food Delivery Platform

Mar 2025 – Apr 2025

Flask, React, PostgreSQL, Redis, Socket.IO, Stripe, Docker, Render

- Designed multi-role REST APIs using Flask Blueprints with JWT authentication and isolated role-based access control for customers, restaurant owners, delivery agents, and admins.
- Implemented real-time order tracking using Socket.IO with Redis as the pub/sub broker, enabling live status updates and instant notifications across distributed clients.
- Integrated Stripe PaymentIntents with idempotency safeguards and signed webhook verification to ensure reliable, fault-tolerant payment workflows.
- Containerized the application stack using Docker Compose with automated Alembic migrations and deployed to **Render** for live access.

Activities

Hackspire Hackathon – Microsoft Learn Student Chapter & ACM Chapter, TIET

- Developed a decentralized civic governance platform for issue reporting and community voting in a competitive field of 80+ participants.

Game Development Workshop – ISTE, TIET

3rd Place

- Designed and shipped a functional game prototype from scratch during a hands-on development workshop.